LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Academic Elective Bidding & Allocation System

Project	Academic Elective Bidding & Allocation System
Project Key	AM278_SE
Student	Soha Rafiq
Laboratory	Lab 2

1. Epics and User Stories

Three major functional themes were created as Epics. The User Stories were organized beneath the appropriate Epic and written from the end-user perspective.

Key	Epic	Description	User Stories
AS-1	Student Elective Bidding	Enables students to view available electives, rank preferences, allocate bidding credits, validate prerequisites, and modify or withdraw bids.	AS-4, AS-6, AS-7, AS-5, AS-8
AS-2	Elective Offering Management	Enables authorized academic staff to manage elective offerings and their associated requirements.	AS-9
AS-3	Elective Allocation & Results	Handles allocation of electives, timetable conflict resolution, and presentation of allocation results.	AS-10, AS-12, AS-11

1.1 User Story Backlog

Key	User Story	Epic	Story Points	Priority
AS-4	View Available Electives	Student Elective Bidding	3	High
AS-6	Allocate Bidding Credits	Student Elective Bidding	5	High
AS-7	Validate Prerequisites	Student Elective Bidding	5	High
AS-5	Rank Elective Preferences	Student Elective Bidding	3	High
AS-8	Modify or Withdraw Bids	Student Elective Bidding	3	Medium
AS-9	Manage Elective Offerings	Elective Offering Management	5	Medium
AS-10	Run Allocation Solver	Elective Allocation & Results	8	High
AS-12	Resolve Timetable Collisions	Elective Allocation & Results	5	High
AS-11	View Allocation Results	Elective Allocation & Results	3	High

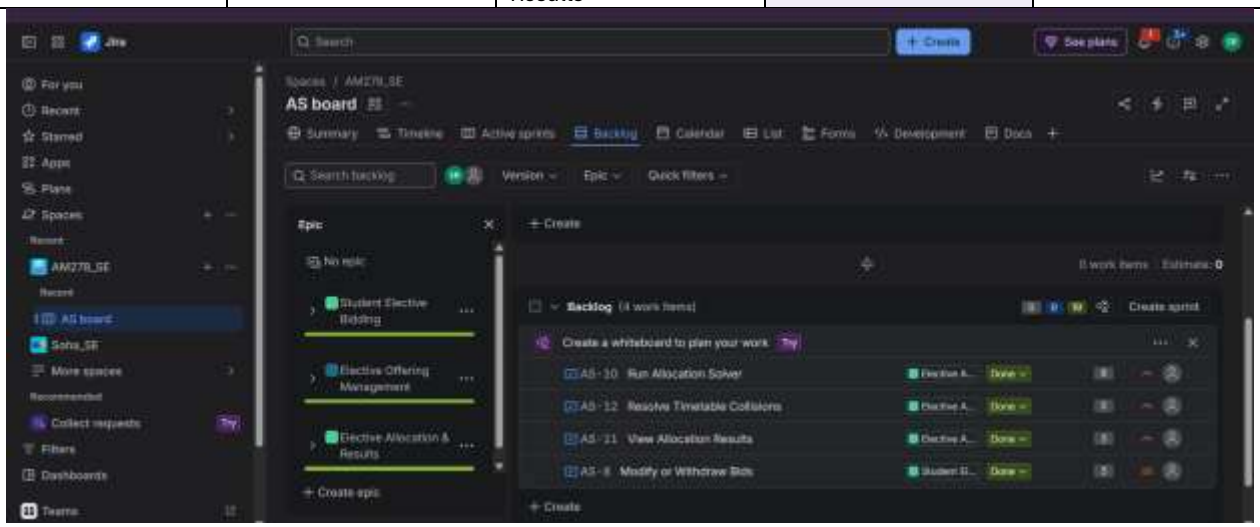


Figure 1. Jira backlog showing the Epics panel and the completed Sprint 2 backlog.

2. Backlog Prioritization and Story-Point Estimation

The backlog was prioritized according to user impact and system importance. Story points were assigned using a Fibonacci-based relative estimation scale. The estimates reflect relative effort, complexity, and uncertainty.

The nine User Stories carry a combined estimate of 40 story points. Sprint 1 was planned with 21 points and Sprint 2 with 19 points.

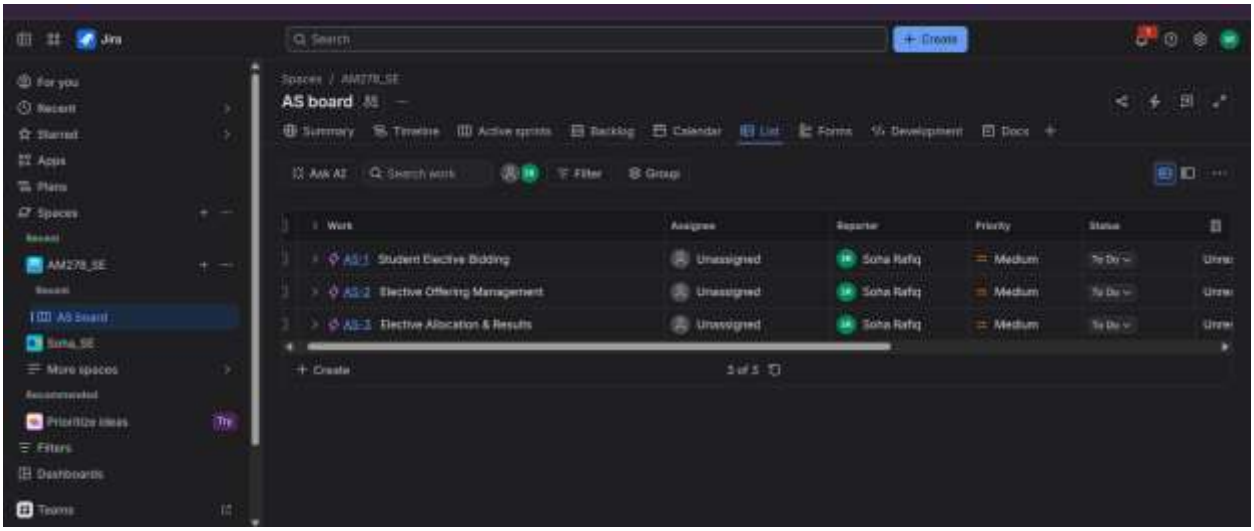


Figure 2. Jira List view showing the three Epics and their current status/priority information.

3. Sprint 1 Simulation

Sprint 1 was created as a one-week sprint. Five User Stories were selected from the backlog, giving a planned total of 21 story points.

Key	Story	Points
AS-4	View Available Electives	3
AS-6	Allocate Bidding Credits	5
AS-7	Validate Prerequisites	5
AS-5	Rank Elective Preferences	3
AS-9	Manage Elective Offerings	5

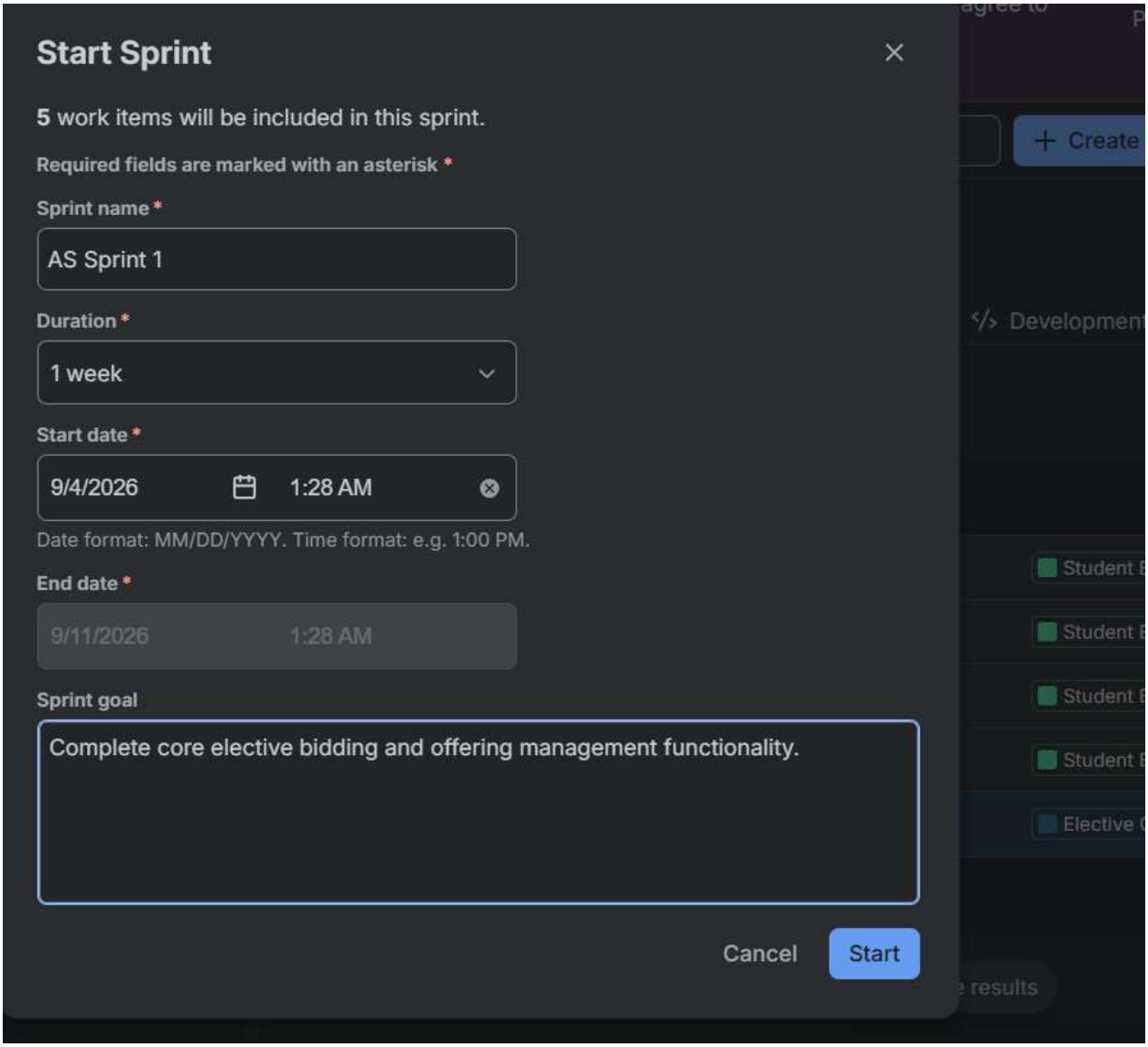


Figure 3. Jira Sprint 1 setup dialog showing five work items and a one-week duration.

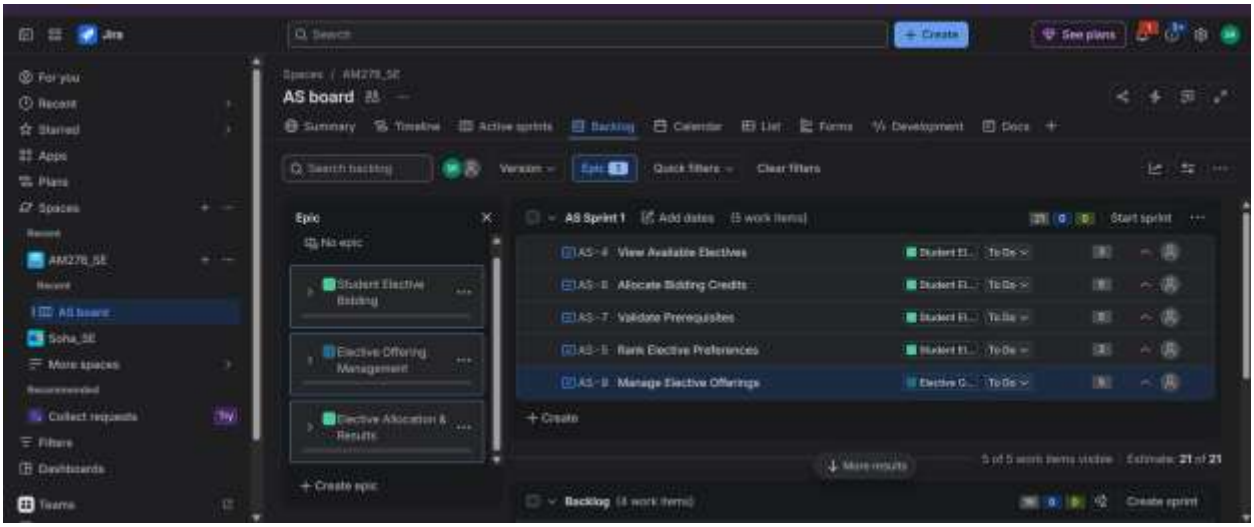


Figure 4. Sprint 1 planned in the Jira backlog with five User Stories and an estimate of 21 points.

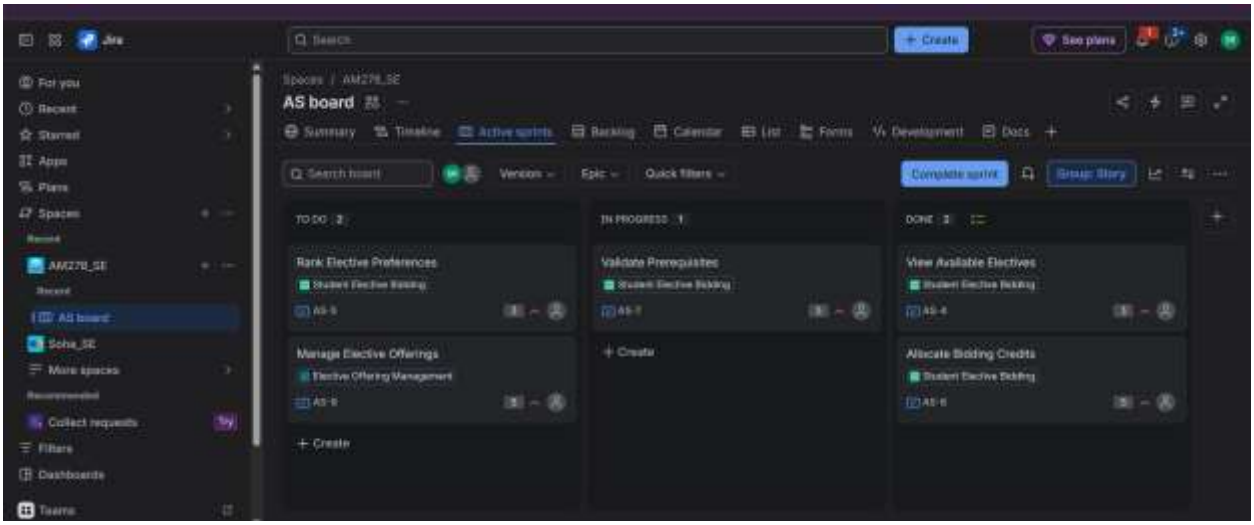


Figure 5. Sprint 1 Active Sprint view showing work progressing through To Do, In Progress, and Done.

4. Sprint 2 Simulation

After Sprint 1, the remaining four User Stories were placed into Sprint 2. Sprint 2 contained 19 story points.

Key	Story	Points
AS-10	Run Allocation Solver	8
AS-12	Resolve Timetable Collisions	5
AS-11	View Allocation Results	3
AS-8	Modify or Withdraw Bids	3

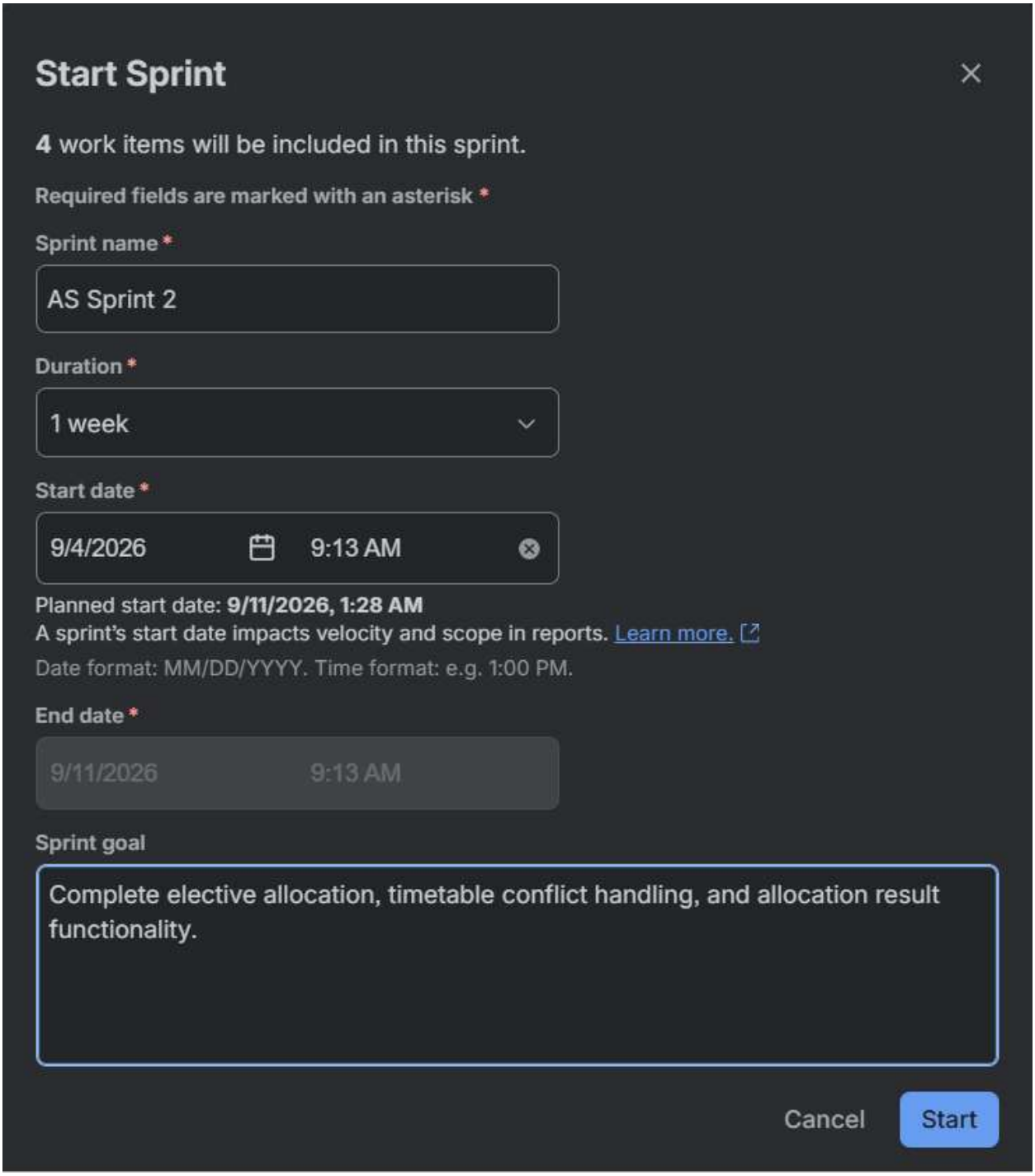


Figure 6. Jira Sprint 2 setup dialog showing four work items and a one-week duration.

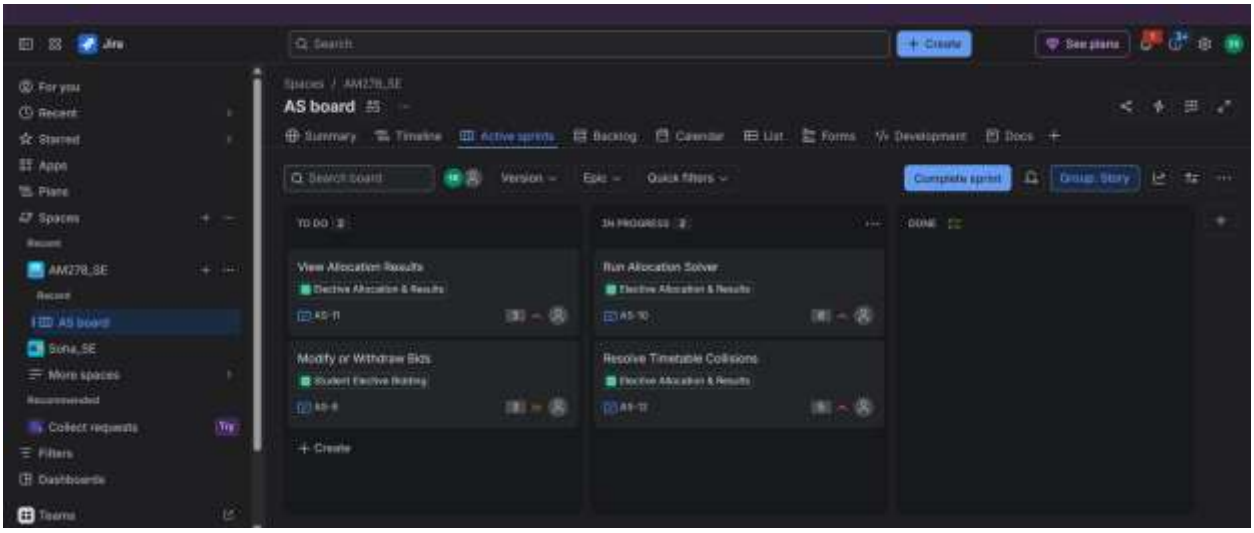


Figure 7. Sprint 2 Active Sprint view showing the remaining stories progressing toward completion.

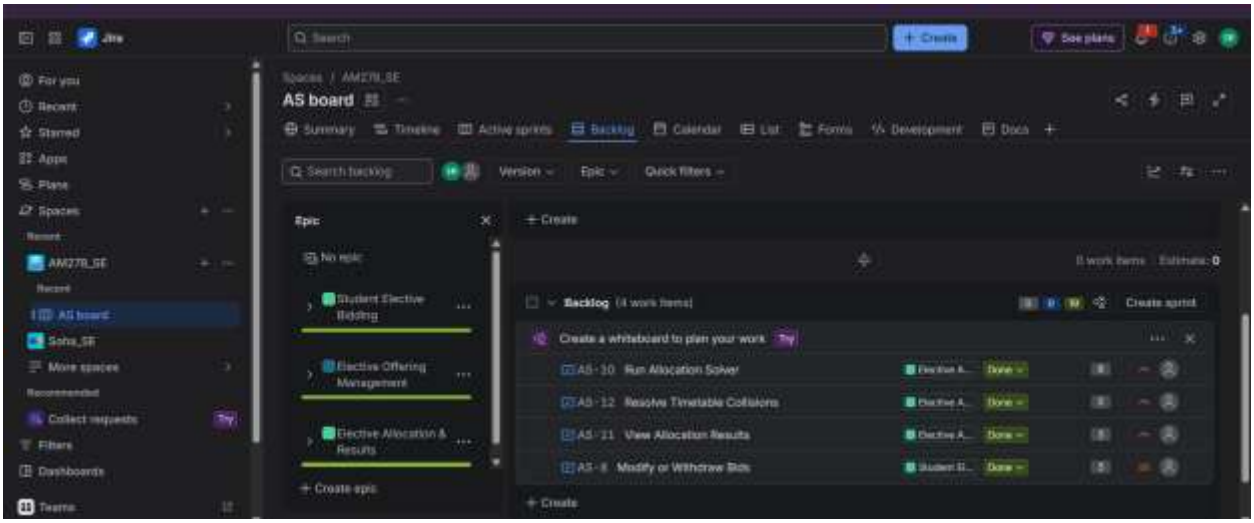


Figure 8. Jira backlog after Sprint 2 showing all four Sprint 2 stories in Done.

5. Timeline and Epic Progress

The Jira Timeline was used to visualize the project Epics and their associated User Stories across September. The completed status of the stories and the relative placement of the Epic work are visible in the timeline.

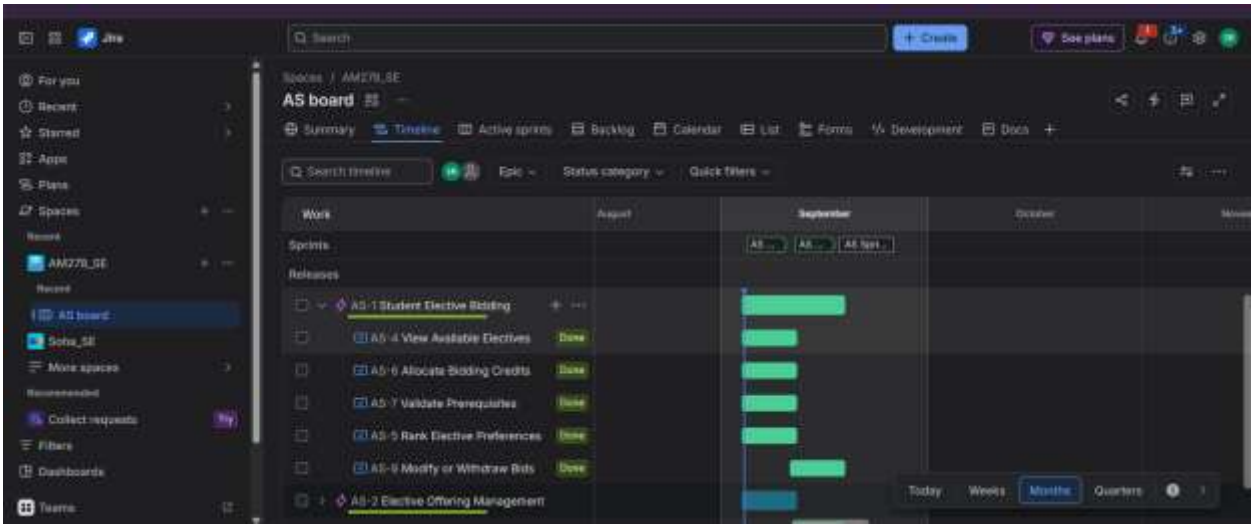


Figure 9. Jira Timeline with the Student Elective Bidding Epic expanded and its User Stories marked Done.

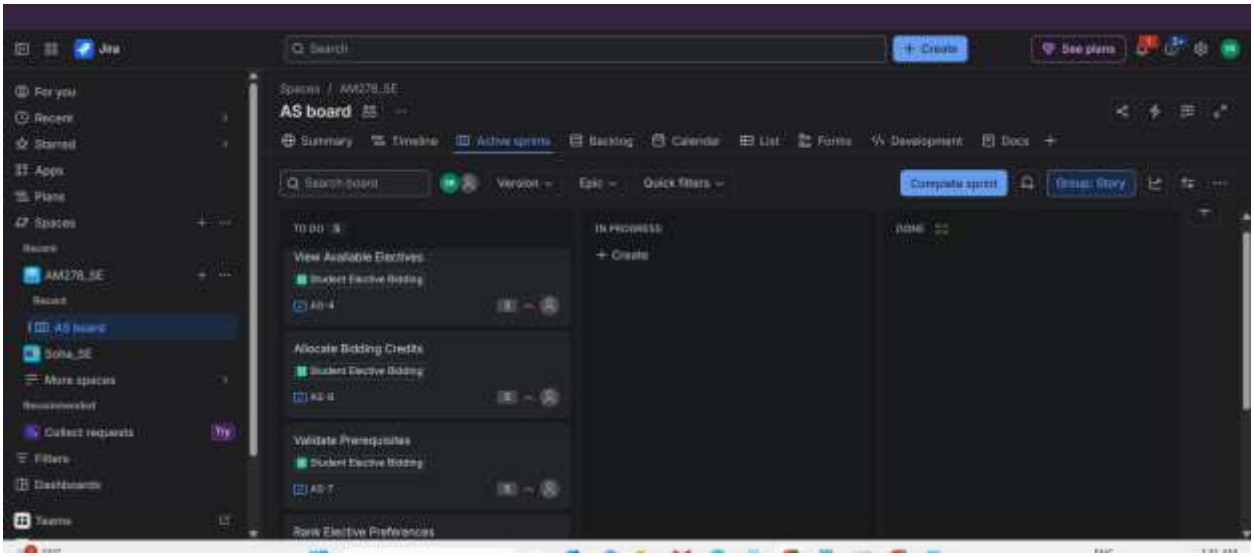


Figure 10. Jira Timeline showing the project work across the Epics.

6. Overall Sprint Completion

The Jira Summary view records nine completed User Stories. The final state demonstrates that all nine User Stories were progressed to Done during the simulated sprint exercise. The three Epics remain visible as the high-level functional structure of the project.

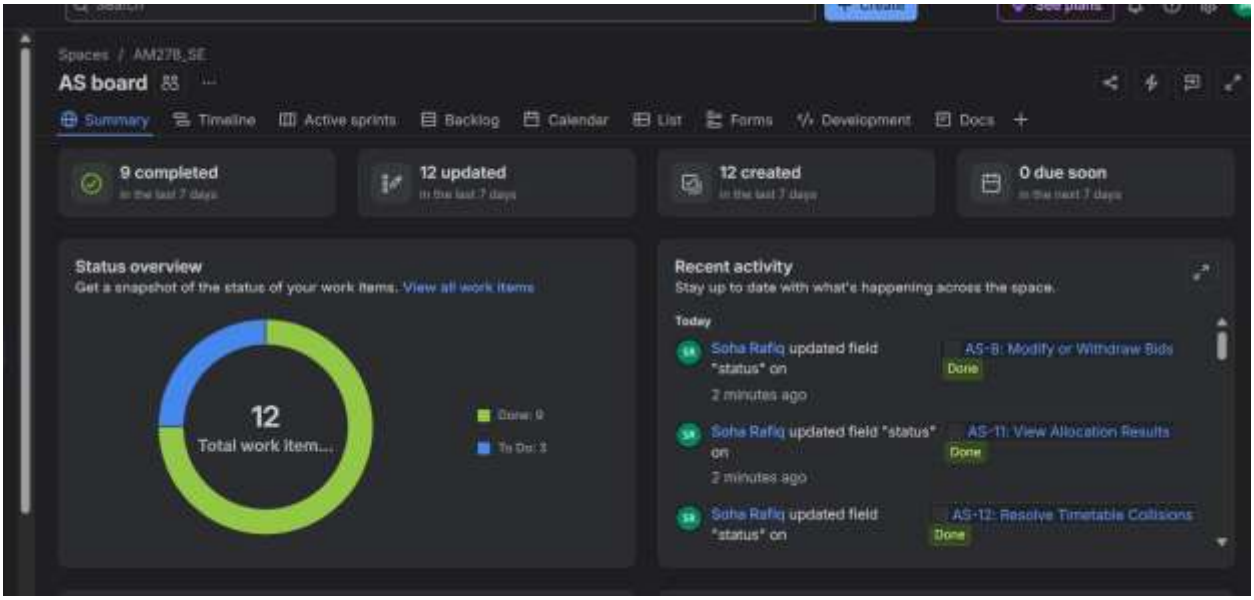
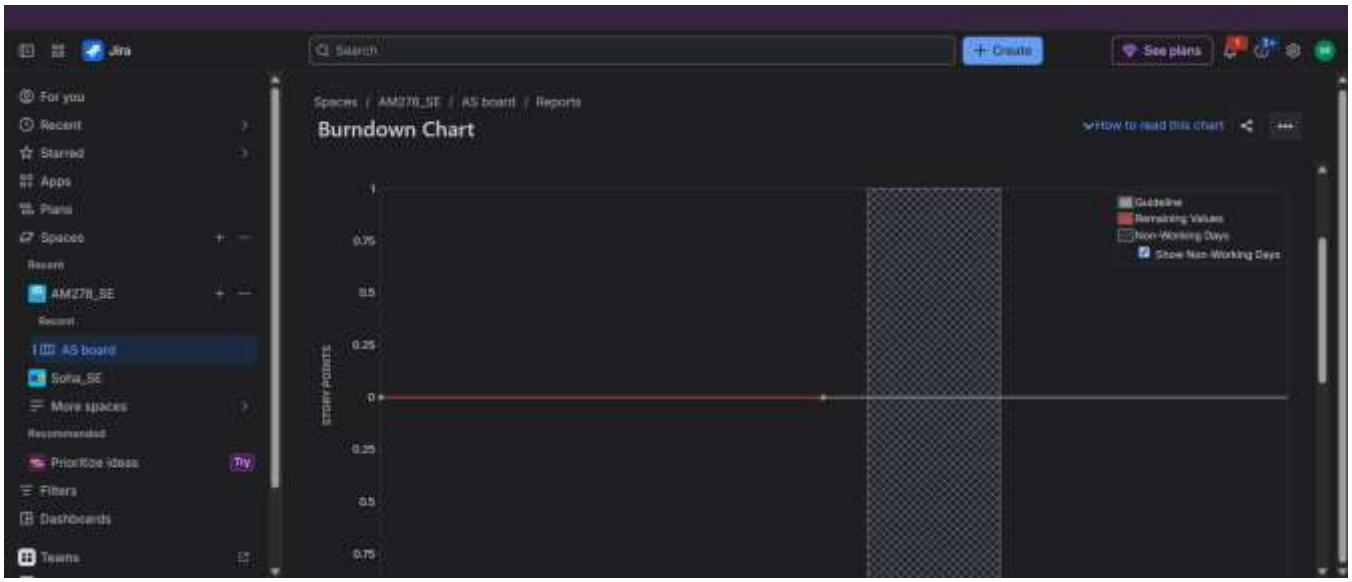


Figure 11. Jira Summary showing 9 completed items and the overall status overview.

7. Burndown Chart



8. Reflection

1. Did your estimations reflect the actual effort?

Overall, the story-point estimates provided a reasonable representation of relative effort. More complex work, such as running the allocation solver, received a higher estimate of 8 points, while simpler work such as viewing allocation results received 3 points. The successful completion of the simulated sprint indicates that the estimates were useful for planning the work.

2. Was your backlog well-prioritized?

Yes. The backlog was organized around the importance of the system functionality. Core student bidding activities were prioritized alongside the supporting offering-management and allocation functionality. The priority values visible in Jira distinguish higher-impact stories from medium-priority supporting work.

3. How did your simulated sprint align with your plan?

The simulation aligned with the planned two-sprint structure. Sprint 1 contained 21 story points and Sprint 2 contained 19 story points. The User Stories were moved through the Scrum workflow and the final Jira Summary shows nine completed items.

4. What insights did the burndown chart give about your team's capacity?

This answer should be finalized after inserting the actual Burndown Chart screenshot. The chart should be used to comment specifically on whether work was completed steadily, whether there were delays or early completions, and whether the team's actual completion pattern was consistent with the planned capacity.

11. Conclusion

The Academic Elective Bidding & Allocation System was successfully represented in Jira using three Epics and nine User Stories. The backlog was prioritized and estimated using story points, and the work was simulated across two one-week sprints. Jira evidence shows the stories progressing through the Scrum workflow and ultimately reaching Done. The remaining submission evidence to insert is the Burndown Chart screenshot and its chart-specific reflection.